

SHUBHA TANAYA

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Education

Carnegie Mellon University

Master of Science in Robotic Systems Development (MRSD) | GPA: 4.0/4.0

Pittsburgh, PA

May 2027

Relevant Coursework: SLAM; Visual Learning & Recognition; Computer Vision, Robot Autonomy, Introduction to Deep Learning

Indian Institute of Technology (IIT) Ropar

Bachelor of Technology in Mechanical Engineering | GPA: 8.31/10.0

Ropar, India

May 2025

Relevant Coursework: Robotic Manipulators, Deep Learning of Physical Systems, Multibody Dynamics

Professional Experience

Komatsu

Robotics Software Engineer Intern | Controls & Automation Team

Warrendale, PA

May 2026 – Aug 2026

- Implemented IMU-free 4D radar odometry with robust dynamic filtering and RCS-weighted matching, improving point cloud quality by 67% in LiDAR-degraded instances caused by high dust in GPS-denied underground mines.
- Designed a semantic ego filter using PCL, applying PCA-based feature extraction to remove self-occlusion and moving vehicle structure from LiDAR point cloud data, improving localization accuracy by 12%.
- Built a real-time camera calibration pipeline in ROS 2 with on-demand image capture and automated intrinsic parameters computation, eliminating a manual, offline calibration step and preventing lens distortion from degrading perception accuracy.

Robotics Institute, Carnegie Mellon University

Graduate Research Assistant | CMU Center for Autonomous Vehicle Research

Pittsburgh, PA

Oct 2025 – Apr 2026

- Mapped the AI and humanoid robotics technology landscape for BMW Group across manufacturing, engineering, and logistics, benchmarking hardware capability and deployment readiness against TRL criteria to inform near-term technology investment decisions.

Academic Projects

Flying Hand — Aerial Manipulator for Non-Destructive Testing

MRSD Capstone Project | AirLab | LeCAR Lab

Carnegie Mellon University

Aug 2025 – Present

- Developing a fully actuated aerial manipulation platform with a base-mounted 6-DOF robotic arm, coordinating drone position-hold with real-time arm target tracking and reactive drone compensation for autonomous NDT in hard-to-reach structures.
- Integrated cuVSLAM for GPS-denied localization on the Jetson Orin Nano's edge compute with PX4 via MAVROS, validating position accuracy against independent RTK-GNSS ground truth to within 1% over a 25 m trajectory segment.
- Designed and validated a 3-layer nested control stack, integrating visual servoing, wrench-based force control, and position control for autonomous contact-based inspection, achieving stable servoing up to 2.5 m standoff and contact error $< 0.15\text{m}$.

Monocular Visual SLAM Robustness Study

SLAM: Course Project | Dr. Michael Kaess

Carnegie Mellon University

Mar 2026 – May 2026

- Ran 1,760 controlled trials of ORB-SLAM3 across 22 EuRoC/KITTI sequences under Gaussian noise, shot noise, and motion blur at five severity levels, systematically mapping which corruptions cause outright tracking failure versus ATE degradation.
- Uncovered a sharp indoor-outdoor performance gap: outdoor tracking collapsed below 40% success at severity-2 corruption while indoor sequences held above 60% through severity 3, with outdoor ATE roughly 100x larger.

UGV-UAV Collaboration for Exploratory Mapping

Modular and Intelligent Robotics Lab | Dr. Ekta Singla

Indian Institute of Technology Ropar

Aug 2024 – May 2025

- Developed high-fidelity digital twins for heterogeneous robots using ROS2 Humble and PX4/Gazebo, synchronizing UGV/UAV dynamics via Micro XRCE-DDS to streamline multi-agent coordination.
- Architected an autonomous navigation stack integrating Cartographer SLAM for UGV mapping and Nav2 planner for dynamic obstacle avoidance, alongside PID-based waypoint following for UAV precision landing.
- Executed hardware-in-the-loop (HIL) validation on Turtlebot3 and Holybro X500 V2, integrating LiDAR and RGB-D sensor fusion to achieve real-time 3D occupancy grid mapping in unknown environments.

Additional Robotics Experience

Safety21 Consortium | F1Tenth Autonomous Racing

Sept 2025 – Nov 2025

- Deployed LiDAR SLAM and particle filter state estimation on a Jetson Orin NX for F1Tenth racing platform, maintaining stable pose tracking at speeds up to 6 m/s and lateral accelerations of 2 m/s^2 during cornering; demoed live at 2025 Safety21 Consortium.

Automotive Club, IIT Ropar | Secretary

Aug 2022 – May 2025

- Led end-to-end club operations and a flagship autonomous navigation project, developing a ROS2 mobile robot in Gazebo with LiDAR SLAM and Nav2, then deploying the stack on TurtleBot3 for a live demonstration at IIT Ropar's annual research symposium.

Skills

Languages & Frameworks: Python, C++, ROS2, PyTorch, AI Agents, PX4 Autopilot, OpenCV, PCL, MAVROS

Localization & Perception: VIO, ORB-SLAM3, Cartographer, Factor Graph Optimization, EKF2, Sensor Modalities (LiDAR, Radar, Camera, GPS, IMU), Point Cloud Processing, ICP, Sensor Calibration, Odometry, Visual Servoing, 3D Reconstruction

Simulation & DevOps: Gazebo, Isaac Sim, MuJoCo, Unreal Engine, RViz, Foxglove, gprMax, Docker, Git, Azure DevOps

Platforms & Hardware: NVIDIA Jetson Orin Nano, Raspberry Pi, Arduino, Linux